

Computer Networks

VLAN Trunking Protocol (VTP) Configuration

elqabasy/gu/network/Assingment1

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VTP Domain: elqabasy.github.io

VTP Password: M@hr0s

1. Project Overview

This project demonstrates the configuration and verification of VLANs using **VLAN Trunking Protocol (VTP)** in a small enterprise network. The network consists of three switches and a router, connected as follows:

R1 <-> SW0-CORE-MAHROS <-> SW1-ACCESS-MAHROS <-> SW2-ACCESS-MAHROS

- **SW0:** VTP Server
 - **SW1:** VTP Client
 - **SW2:** VTP Client
 - VLANs: 10 (IT Department), 20 (HR Department)
 - Inter-VLAN routing is performed using **Router-on-a-Stick** with R1.
-

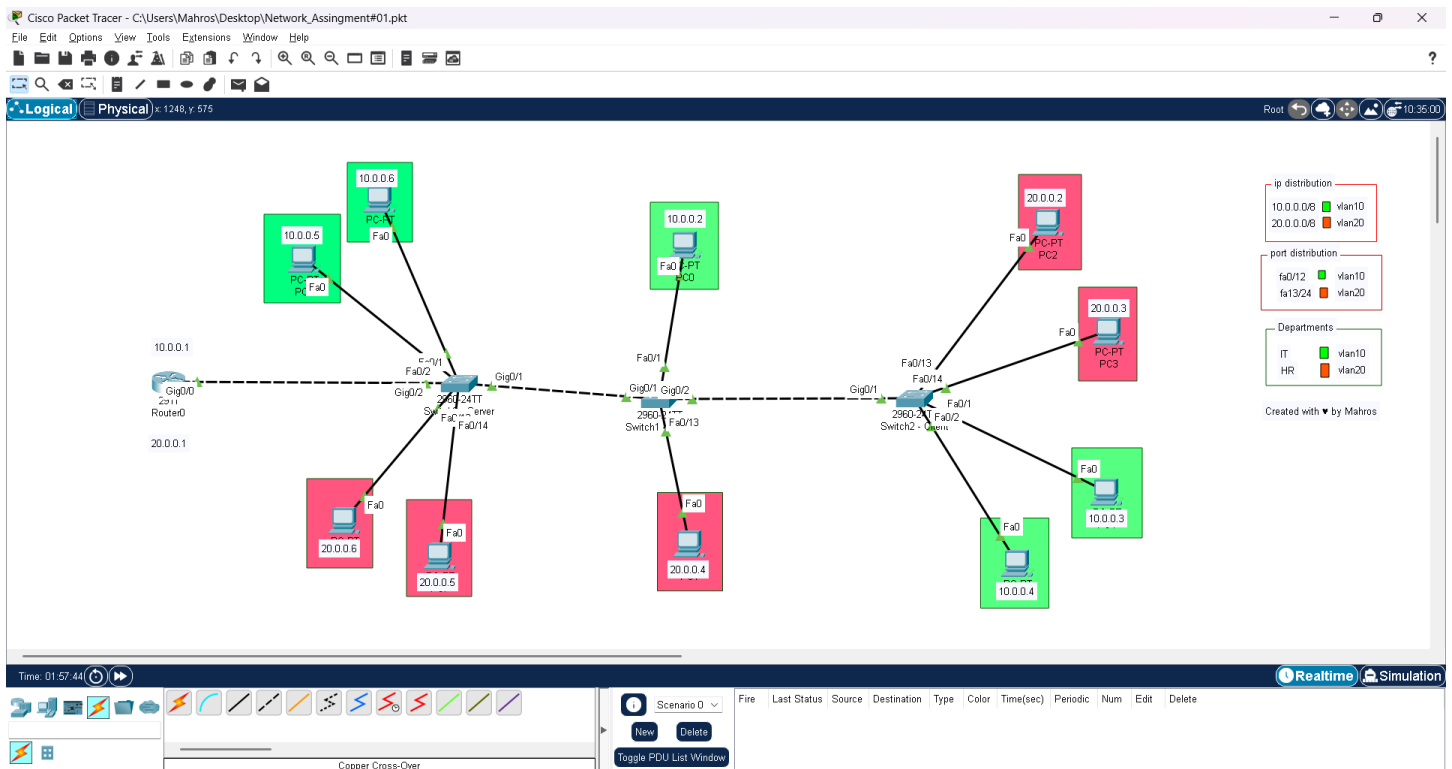
2. Topology Overview

All devices and connections were created in **Packet Tracer**. The network is structured to ensure proper VLAN isolation, trunking between switches, and inter-VLAN routing.

Connections:

Device 1	Port	Device 2	Port
R1	Fa0/0	SW0	Gi0/2
SW0	Gi0/1	SW1	Gi0/1
SW1	Gi0/2	SW2	Gi0/1

Screenshots:



3. VTP Server Configuration (SW0-CORE-MAHROS)

Objective: Configure SW0 as VTP Server to propagate VLANs automatically.

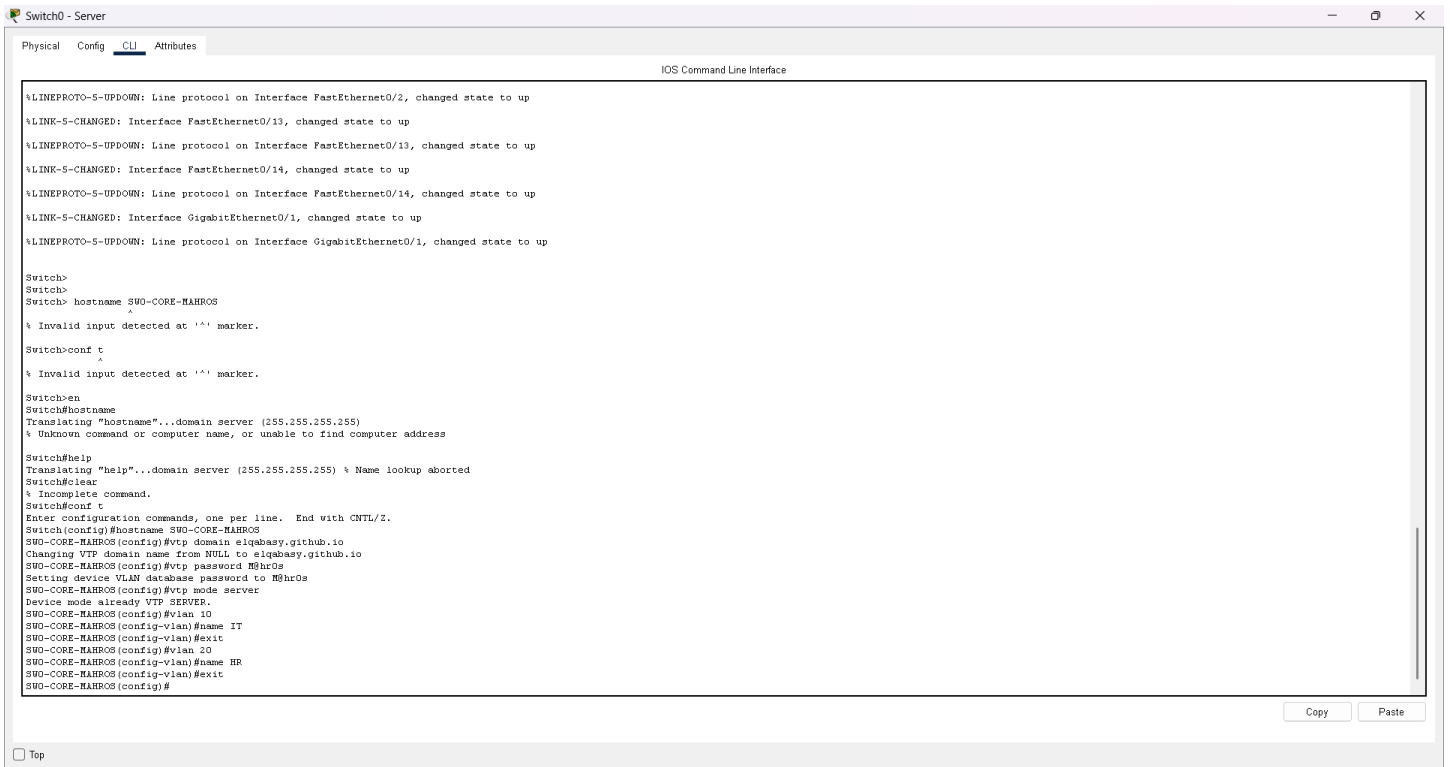
Commands Used:

```
hostname SW0-CORE-MAHROS
vtp domain elqabasy.github.io
vtp password M@hr0s
vtp mode server
```

```
vlan 10
 name IT
exit
```

```
vlan 20
 name HR
exit
```

Screenshot of VLAN creation:



Physical

Config

CLI

Attributes

IOS Command Line Interface

Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/13, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/13, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/14, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/14, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Switch>

Switch>

Switch>en

Switch#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)# hostname SW2-ACCESS-MAHROS

SW2-ACCESS-MAHROS(config)#vtp domain elqabasy.github.io

Changing VTP domain name from NULL to elqabasy.github.io

SW2-ACCESS-MAHROS(config)#vtp password M@hr0s

Setting device VLAN database password to M@hr0s

SW2-ACCESS-MAHROS(config)#vtp mode client

Setting device to VTP CLIENT mode.

SW2-ACCESS-MAHROS(config)#

Copy

Paste

☐ Top


```
vtp password M@hr0s
vtp mode client
```

Screenshot of VTP Status Verification:

The screenshot shows a web-based CLI interface for a switch named 'Switch1 - Client'. The 'CLI' tab is selected, displaying the 'IOS Command Line Interface'. The interface shows the following text:

```
SW1-ACCESS-MAHROS con0 is now available

Press RETURN to get started.
```

Below this, the user has entered the command `show vlan brief`, and the output is displayed as follows:

```
SW1-ACCESS-MAHROS>en
SW1-ACCESS-MAHROS#show vlan
SW1-ACCESS-MAHROS#show vlan bri
SW1-ACCESS-MAHROS#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	
10 IT	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12
20 HR	active	Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

The interface also includes a 'Copy' button and a 'Paste' button at the bottom right, and a 'Top' link at the bottom left.

Switch2 - Client

Physical
Config
CLI
Attributes

IOS Command Line Interface

SW2-ACCESS-MAHROS con0 is now available

Press RETURN to get started.

SW2-ACCESS-MAHROS>en
SW2-ACCESS-MAHROS#show vlan br
SW2-ACCESS-MAHROS#show vlan brief

VLAN Name	Status	Ports
1 default	active	Gig0/2
10 IT	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12
20 HR	active	Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

SW2-ACCESS-MAHROS#

Copy
Paste

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5. Trunk Configuration

Objective: Configure trunk links between switches to carry VLAN traffic and propagate VTP updates.

Commands Example (SW0 to SW1):

```
interface gi0/1
switchport mode trunk
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,20
description Trunk_to_SW1_Configured_by_mahros
```

Other Trunks:

- SW1 Gi0/2 -> SW2 Gi0/1

- SW0 Gi0/2 -> R1 Fa0/0

Screenshot of Trunk Verification:

Switch1 - Client

Physical
Config
CLI
Attributes

IOS Command Line Interface

```

SW1-ACCESS-MAHROS#show vl
SW1-ACCESS-MAHROS#show vlan ?
  brief  VTP all VLAN status in brief
  id      VTP VLAN status by VLAN id
  name    VTP VLAN status by VLAN name
  <cr>
SW1-ACCESS-MAHROS#show vlan br
SW1-ACCESS-MAHROS#show vlan brief

```

VLAN Name	Status	Ports
1 default	active	
10 IT	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12
20 HR	active	Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```

SW1-ACCESS-MAHROS#show inter
SW1-ACCESS-MAHROS#show interfaces tru
SW1-ACCESS-MAHROS#show interfaces trunk

```

Port	Mode	Encapsulation	Status	Native vlan
Gig0/1	on	802.1q	trunking	1
Gig0/2	on	802.1q	trunking	1

```

Port
Gig0/1
Gig0/2

```

Port	Vlans allowed on trunk
Gig0/1	10,20
Gig0/2	10,20

```

Port
Gig0/1
Gig0/2

```

Port	Vlans allowed and active in management domain
Gig0/1	10,20
Gig0/2	10,20

```

Port
Gig0/1
Gig0/2

```

Port	Vlans in spanning tree forwarding state and not pruned
Gig0/1	10,20
Gig0/2	10,20

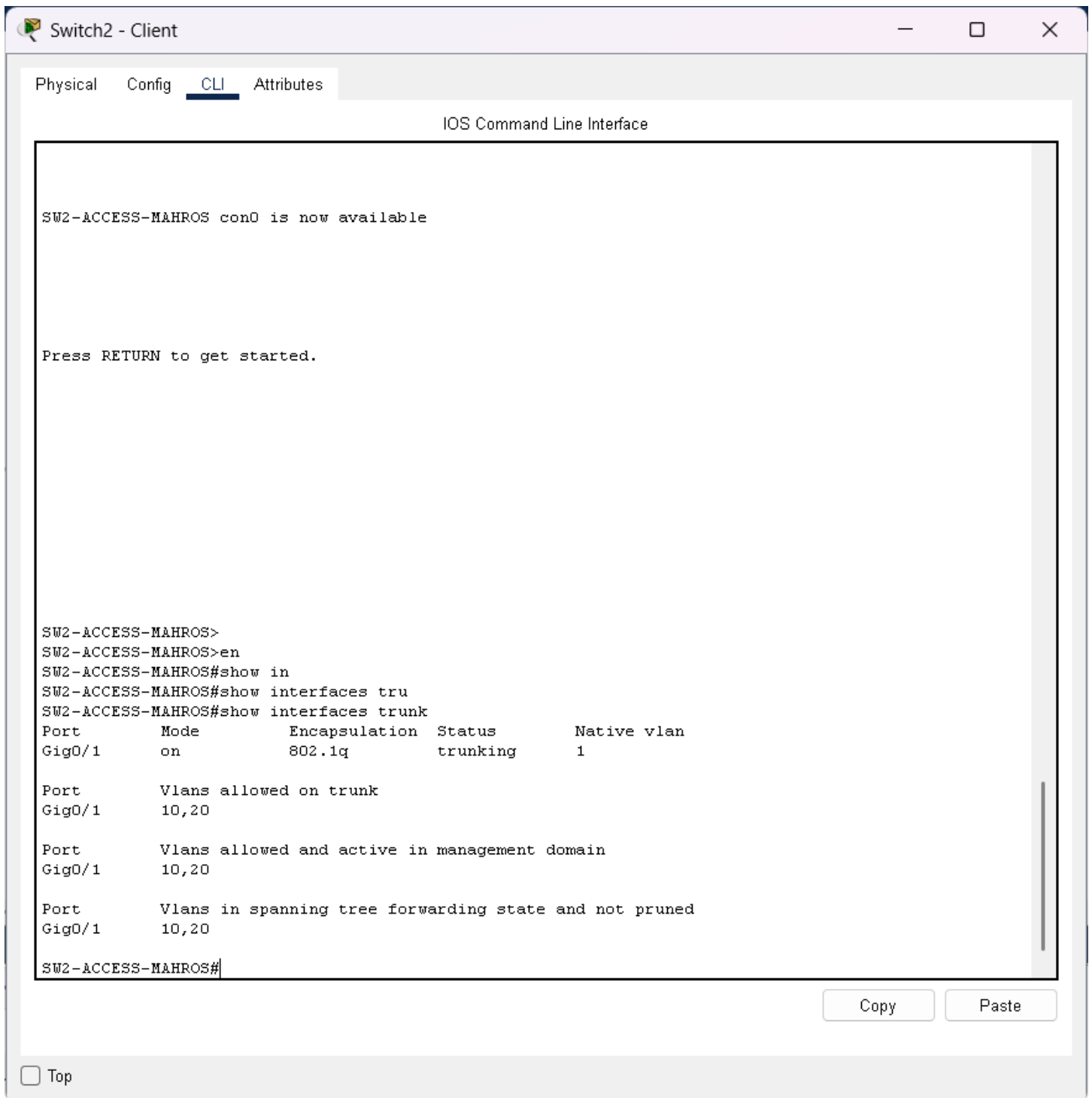
```

SW1-ACCESS-MAHROS#

```

Copy
Paste

☐ Top



6. Access Port VLAN Assignment

Objective: Assign access ports to VLANs according to department.

VLAN Mapping:

- **VLAN 10 (IT Department):** Fa0/1 – Fa0/12
- **VLAN 20 (HR Department):** Fa0/13 – Fa0/24

Commands Example (SW0):

```
interface range fa0/1 - 12
switchport mode access
switchport access vlan 10
description IT_Department_PC_Port_Configured_by_mahros
```

```

interface range fa0/13 - 24
  switchport mode access
  switchport access vlan 20
  description HR_Department_PC_Port_Configured_by_mahros

```

Screenshot of VLAN Assignment:

7. IP Addressing

VLAN 10 – IT Department

PC	IP Address	Subnet Mask	Default Gateway
PC0	10.0.0.2	255.0.0.0	10.0.0.1
PC1	10.0.0.3	255.0.0.0	10.0.0.1

VLAN 20 – HR Department

PC	IP Address	Subnet Mask	Default Gateway
PC2	20.0.0.2	255.0.0.0	20.0.0.1

Screenshot of PC IP Configuration:

8. VLAN Communication Testing

Intra-VLAN (Ping within VLANs):

- PC0 -> PC1 (VLAN 10) -> Success
- PC2 -> PC2 (VLAN 20) -> Success

Inter-VLAN (Ping across VLANs before router):

- PC0 -> PC2 -> Fail (as expected due to VLAN isolation)

Screenshot of Ping Test:

```

PC2
Physical Config Desktop Programming Attributes
Command Prompt
Control-C
^C
C:\>ping 20.0.0.1

Pinging 20.0.0.1 with 32 bytes of data:

Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time=9ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 20.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>ping 20.0.0.2

Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time=3ms TTL=128
Reply from 20.0.0.2: bytes=32 time=6ms TTL=128
Reply from 20.0.0.2: bytes=32 time=7ms TTL=128
Reply from 20.0.0.2: bytes=32 time=7ms TTL=128

Ping statistics for 20.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 7ms, Average = 5ms

C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...: FE80::200:CFF:FE8:A47E
    IPv6 Address...: ::
    IPv4 Address...: 20.0.0.2
    Subnet Mask...: 255.0.0.0
    Default Gateway...: ::
                        20.0.0.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address...: ::
    IPv6 Address...: ::
    IPv4 Address...: 0.0.0.0
    Subnet Mask...: 0.0.0.0
    Default Gateway...: ::
                        0.0.0.0

C:\>

```

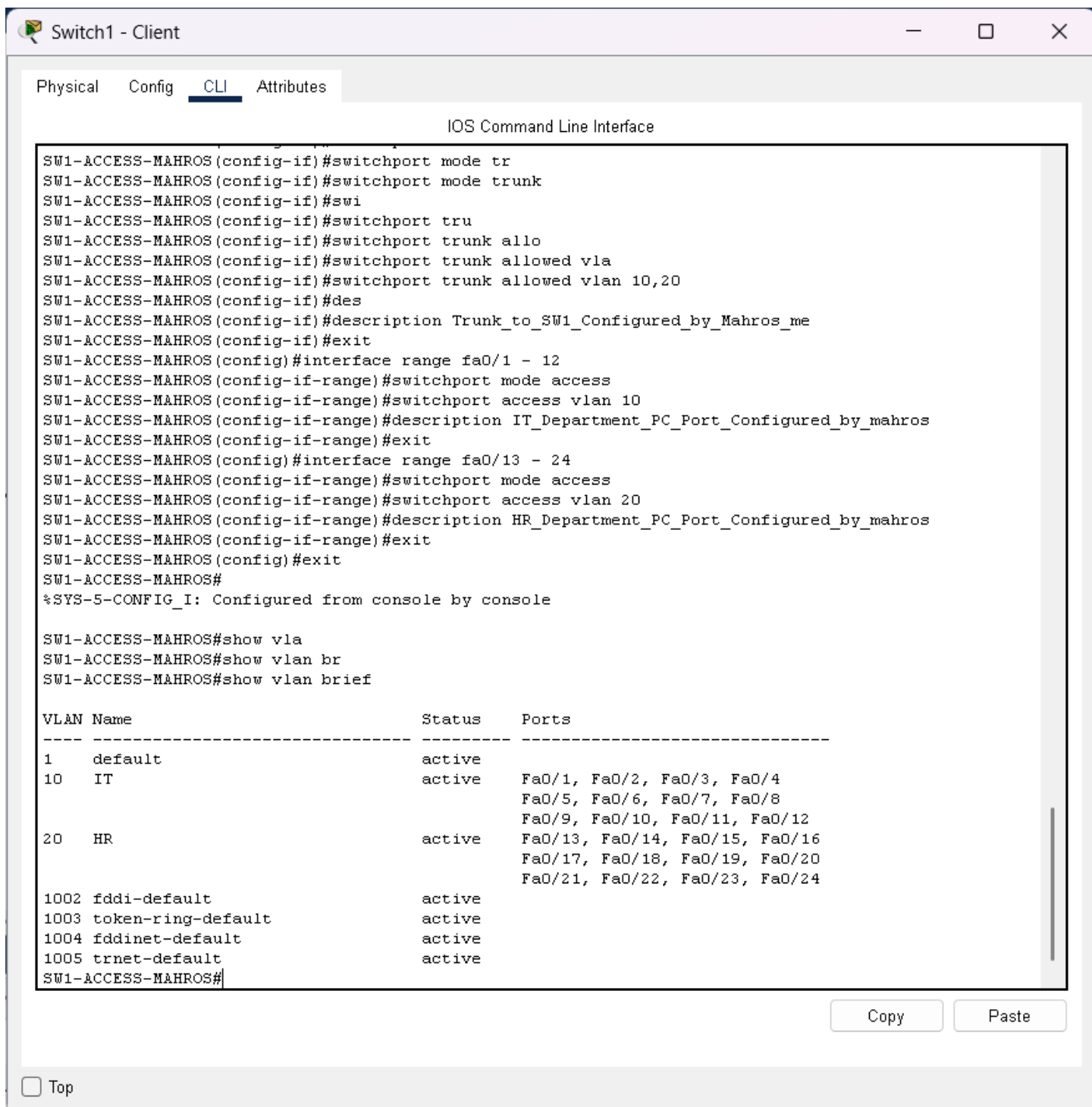
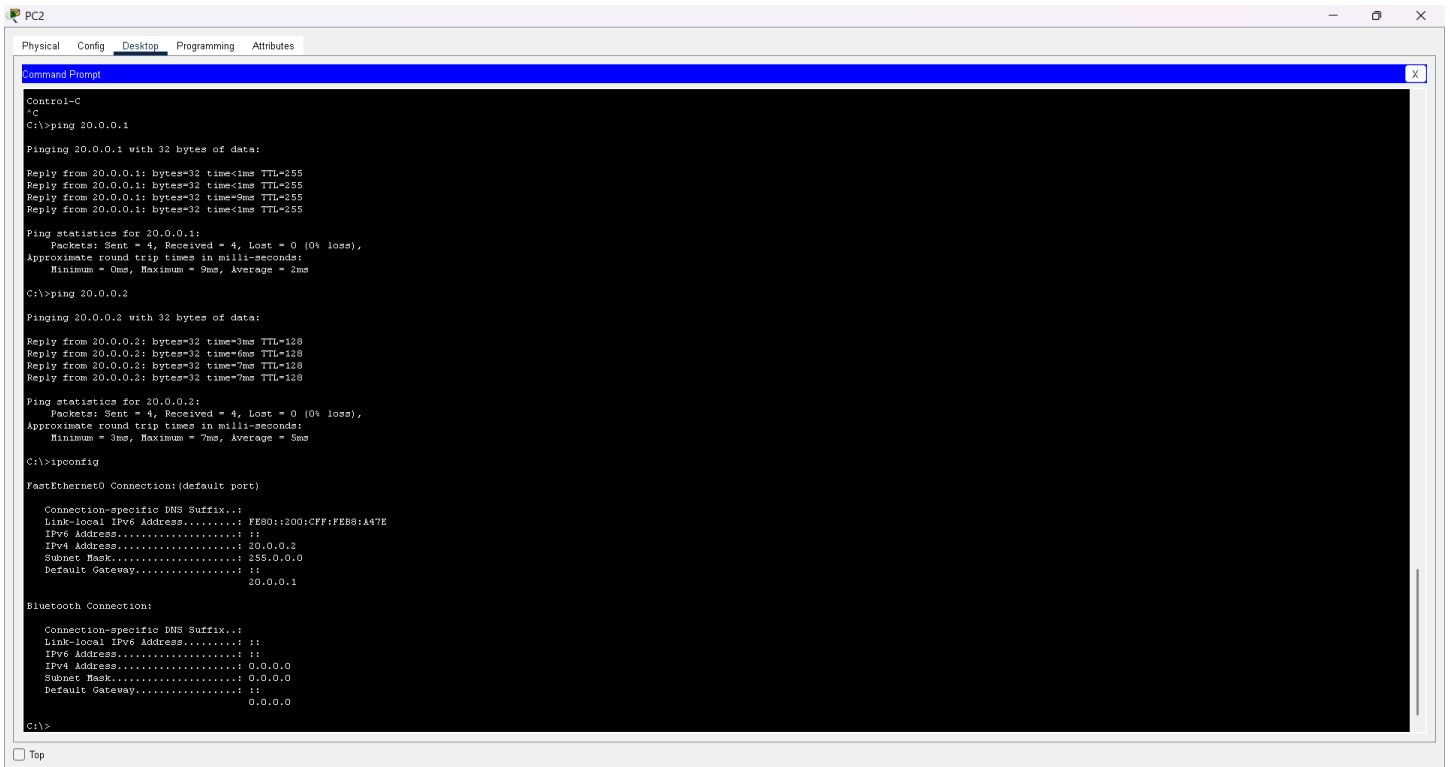


Figure 1: VLAN Assignment



```
PC2
Physical Config Desktop Programming Attributes
Command Prompt
Control-C
^C
C:\>ping 20.0.0.1

Pinging 20.0.0.1 with 32 bytes of data:

Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time=9ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 20.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>ping 20.0.0.2

Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time=3ms TTL=128
Reply from 20.0.0.2: bytes=32 time=6ms TTL=128
Reply from 20.0.0.2: bytes=32 time=7ms TTL=128
Reply from 20.0.0.2: bytes=32 time=7ms TTL=128

Ping statistics for 20.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 7ms, Average = 5ms

C:\>ipconfig

FastEthernet0 Connection: (default port)

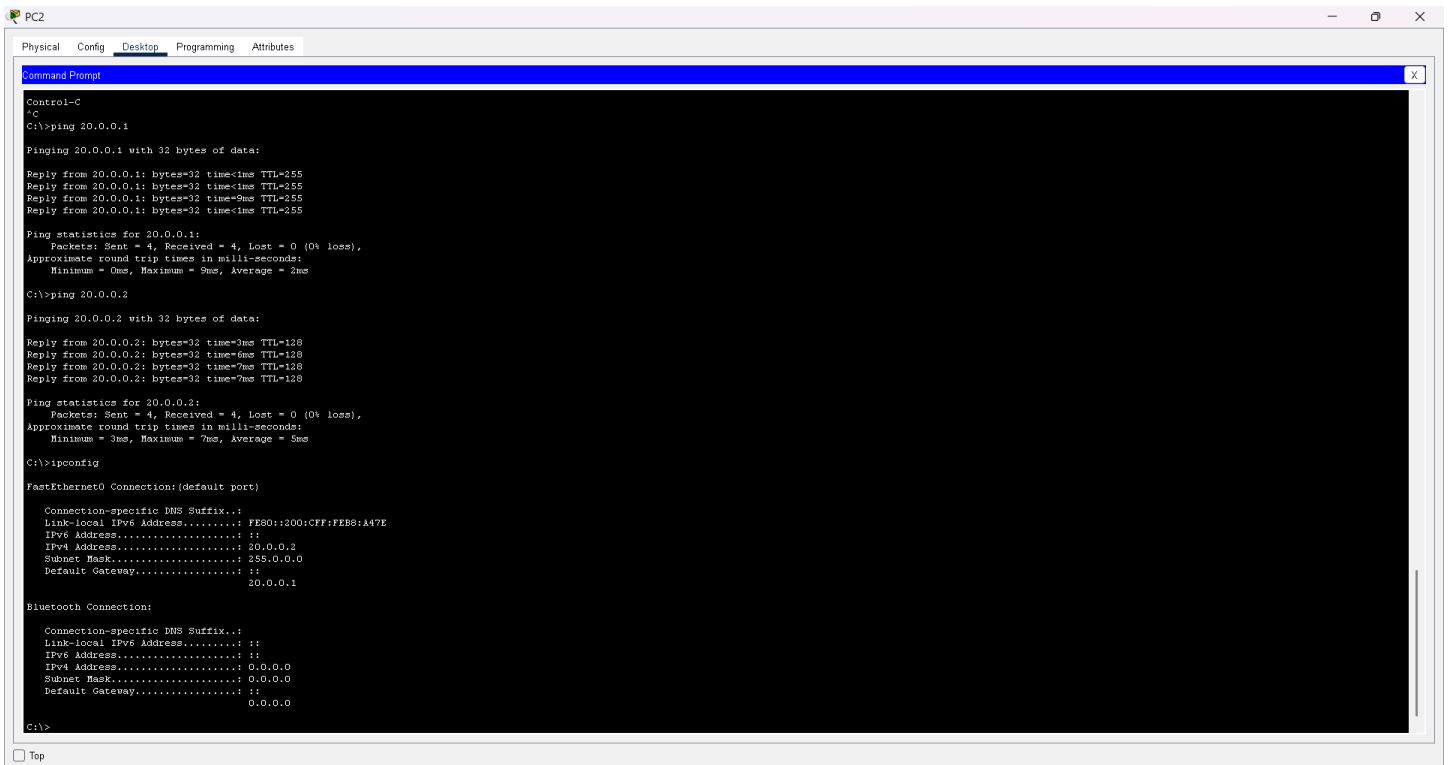
    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::200:CFF:FEB8:A47E
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 20.0.0.2
    Subnet Mask . . . . .: 255.0.0.0
    Default Gateway . . . . .: ::
                                   20.0.0.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>
```

Figure 2: PC IP Config



```
PC2
Physical Config Desktop Programming Attributes
Command Prompt
Control-C
^C
C:\>ping 20.0.0.1

Pinging 20.0.0.1 with 32 bytes of data:

Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255
Reply from 20.0.0.1: bytes=32 time=9ms TTL=255
Reply from 20.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 20.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>ping 20.0.0.2

Pinging 20.0.0.2 with 32 bytes of data:

Reply from 20.0.0.2: bytes=32 time=3ms TTL=128
Reply from 20.0.0.2: bytes=32 time=6ms TTL=128
Reply from 20.0.0.2: bytes=32 time=7ms TTL=128
Reply from 20.0.0.2: bytes=32 time=7ms TTL=128

Ping statistics for 20.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 7ms, Average = 5ms

C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::200:CFF:FEB8:A47E
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 20.0.0.2
    Subnet Mask . . . . .: 255.0.0.0
    Default Gateway . . . . .: ::
                                   20.0.0.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>
```

9. Router-on-a-Stick Configuration (R1-MAHROS)

Objective: Enable inter-VLAN routing to allow communication between VLAN 10 and VLAN 20.

Router Subinterface Configuration:

```
interface fa0/0.10
```

```
encapsulation dot1Q 10
ip address 10.0.0.1 255.0.0.0
description VLAN10_IT_MAHROS
```

```
interface fa0/0.20
encapsulation dot1Q 20
ip address 20.0.0.1 255.0.0.0
description VLAN20_HR_MAHROS
```

```
interface fa0/0
no shutdown
```

Update PC Default Gateways:

- VLAN 10 PCs -> 10.0.0.1
- VLAN 20 PCs -> 20.0.0.1

Screenshot of Router Interfaces:

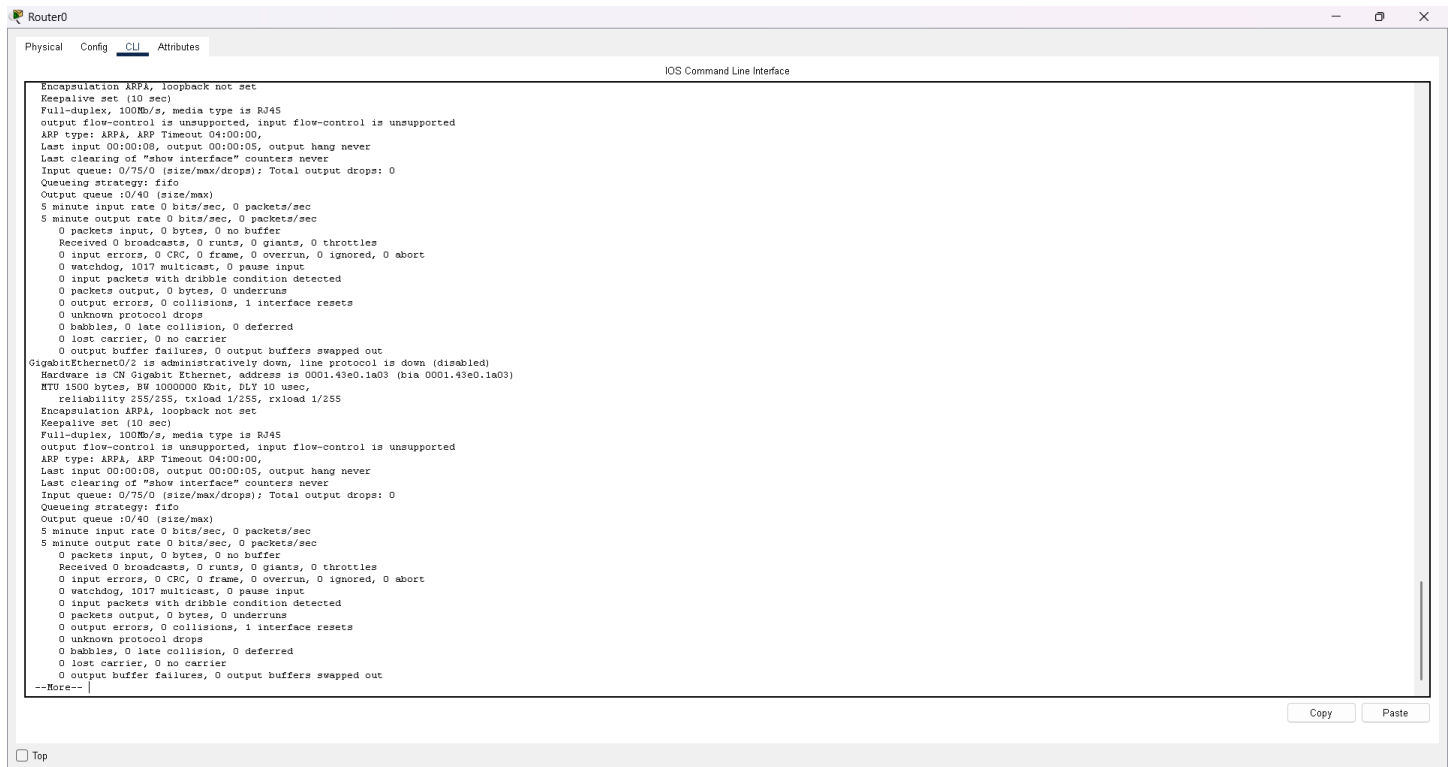


Figure 3: Router Interfaces

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